

Figure 14.9 Unshielded Twisted Pair Cable

It consists of two insulating copper wires (1mm thick). The wires are twisted together in a helical form to reduce electrical interference from similar pair.

Advantages

- It has high speed capacity.
- Higher grades of UTP are used in LAN technologies like Ethernet.

Shielded Twisted Pair Cable

This cable has a metal foil or braided-mesh. Electromagnetic noise penetration is prevented by metal casing. Shielding also eliminates crosstalk. It is faster than unshielded and coaxial cable.

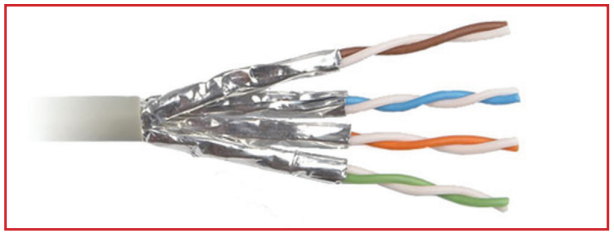


Figure 14.10 Shielded Twisted Pair

Advantages

- It can be used for Analog or Digital transmission
- It increases the signalling rate.
- It eliminates crosstalk.

Coaxial Cable

- Coaxial is called by this name because it contains two conductors that are parallel to each other.
- Copper is used in this as a central conductor. It is surrounded by PVC installation.
- Outer metallic wrapping is used as a shield against noise and as a second conductor which completes the circuit.
- The outer conductor is also encased in an insulating sheath. The outermost part is the plastic cover which protects the whole cable.
- The most common coaxial standards are:
  - 50-Ohm RG-7 or RG-11 : used with thick Ethernet.
  - 50-Ohm RG-58 : used with thin Ethernet
  - 75-Ohm RG-59 : used with cable television
  - 93-Ohm RG-62 : used with ARCNET.

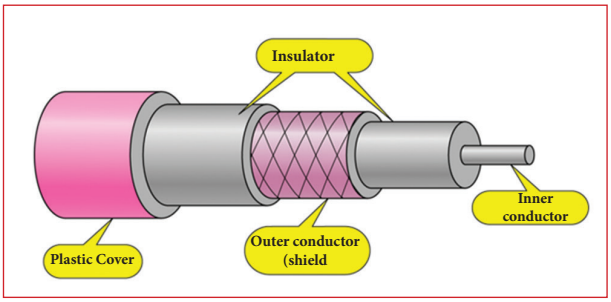


Figure 14.11 Coaxial Cable

Advantages

- It is used in long distance telephone lines.
- It transmits digital signals at a very high rate of 10Mbps.
- It has higher noise immunity.
- It transmits data without distortion.